

ActInf GuestStream 049.1 ~ "Clickbait, consciousness science, and responsible journalism"

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hello and welcome it's July 26th 2023 we're here in octave inference live stream for 9.1 with Megan Peters and Nora Bradford so thank you both for joining we're looking forward to a presentation followed by a discussion on clickbait Consciousness science and responsible journalism so back in to you thank you and thanks so much Daniel for having us we're really happy to be here uh and to chat about this so I'll talk for a little over 10 minutes and then we'll hopefully have a nice discussion here um so yeah we're really happy to be here because I think that this is a really topical uh really um yeah very topical thing for us to be talking about right now it's very in the media it's really in response to the way the media has been covering Consciousness science just in the past few weeks uh so here here we go I'll have a few things to share with you okay so since we're talking about Consciousness science um let's start with a provocative question so you are conscious how is this possible how is it possible that your brain creates the conscious experience of you and so this is a captivating and it's a stimulating kind of crazy question it's a really huge question uh borders on the religious and spiritual in many ways it borders on The Cutting Edge of physics in many ways and so the study of this question and the field of Consciousness science more generally it's really straddling the border between the Natural Sciences and philosophy and spirituality in a way that a lot of other domains maybe don't need to do and so as a result this field of Consciousness science it really captures the hearts and and minds and sometimes pretty wild imaginations of all sorts of folks as I said from psychologists and biologists and neuroscientists all the way to computer scientists and Engineers to religious and spiritual leaders basically anybody who wonders about our origins or place in the universe why that we know that we are here why there is something that it is like to be us they are the people who are interested in this question and that's probably a lot of people who are listening and watching right now so it's it's actually really quite easy I think to get swept up in The Sensational nature of this question and Consciousness science is a really big deal it touches on so many aspects of our everyday

experience um and so this kind of leads us to another version of this question which is is this even really science is this something that can be even studied scientifically or does it transcend Beyond on scientific inquiry did Consciousness scientists do something that can be characterized really as science or maybe they're studying something that's a little bit more on the border like I don't know telepathy or pan psychism like maybe rocks and trees are also conscious wouldn't that be crazy or maybe the vibrations of the use mean that the universe is conscious so it it kind of quickly starts to feel a little a little fuzzy a little less scientific um and that's what I want to talk about today is that perception and how that is kind of exacerbated by the way that this field is portrayed in the media and maybe what we can do about it um okay and so all this means that the conscience and science is really difficult not just to do scientifically and it is really difficult I promise you um but it's also really difficult from a social perspective so more than than for some other fields those of us who who do Consciousness science and who study this scientifically we have a really crucial and difficult responsibility which is to accurately and precisely communicate the state of our field and its methods and its discoveries and how it relates to other fields from physics to medicine and and Beyond um and it's in fact so difficult that our field has actually written a number of papers about just how difficult it is so um this is a fairly recent paper always included in many other um and this is just from a few years ago and uh these papers and this one in particular really highlight why our field in in uh specific has kind of an uphill battle compared to a lot of other areas of study especially when it comes to potential misrepresentation in the media so I thought that maybe in order to drive this point home I could actually give you some examples of how subconsciousness science has been misrepresented in the media so that we can get a handle on this we can see it concretely and as I'm giving you some of these examples I want you to to try to imagine how it would sound if the same kinds of statements the same kind of sweeping claims were made about discoveries in biology or chemistry Neuroscience climate science engineering that kind of thing all right so let's do some examples here is the first one okay and this is an example which is actually 13 years old at this point um as well as I think you'll see maybe it hasn't gotten so much better over the past 13 years but in this example the title sizing of Consciousness by its bits we have Christoph Coke who is an expert on Consciousness science at Caltech being quoted as saying I love his ideas it's the only real promising fundamental Theory and here is deciding to present just this one scientist's opinion as fact without challenging it all without presenting any other opinions at all I'll point out that this opinion was actually not representative of the consensus in the field at that time it was this

one scientist's point of view and it's not actually representative of the consensus in the field now either and so I want you to take this quote but I want you to replace Consciousness with curing cancer so this opinion presented as fact is akin to writing that one scientist's theory about a cure for cancer is the only really promising fundamental theory for curing cancer and without presenting any other possibilities that that maybe more work needs to be done or more experiments need to be run this is really hyperbolic and it seems like maybe this kind of hyperbole would be hopefully at least a little less likely to happen in biomedicine or at least with receive more more pushback if it were presented this way so here's another example and this one's a bit more recent so this one's from um 2018. uh just is okay so the electrical activity that moves across neurons is thought by some scientists to contribute to human consciousness this is like a perfectly reasonable statement right like totally fine totally legit but then the journalist goes on to say if electrical activity is being disrupted by anesthetic and plants two causing them to lose Consciousness does that mean in some way that they are conscious okay so first of all these statements are are pretty vague um that's not really super surprising from a popular science piece and it's not really the the primary problem here saying that a liberty that moves across neurons thought to distance from destruction of electrical activity to plants being conscious and this is pretty Sensational kind of to the extreme and it makes this fuzzy border between Consciousness science and pseudoscience which is this border that we are trying so hard to delay all the time it makes it even fuzzier gives us an even harder problem to fight against as Consciousness and it isn't Just Happening 13 years ago five years ago it still has like if I were talking just last uh there was this post which was held in conjunction with the association for the scientific study of Consciousness annual meeting which was in New York City and at this public event Christoph Koch who you can see here on the right and Dave Chalmers here's on the left they presented their 25 year old bet about whether the neural correlates of Consciousness the ones that go up and down with Consciousness goes up and down uh would be found later and during this public event some new results were unveiled for the very first time from one of the ongoing projects which was designed to challenge two of the many theories about how Consciousness arises in the brain and those two theories are called the global neuronal workspace Theory and the integrated information Theory and the hot off the press results that were shown in this event uh they actually ended up challenging both theories they showed that the neural correlates of Consciousness are still unknown that neither theory was really strongly supported by the data that were collected and so on but that's not the story that's not exciting and so that's not the story that the Press told and so here

is what the media actually said okay so first we're going to start with nature nature the journalism side of this Flagship scientific journal you'd hope that they would be pretty careful about their scientific journalism but instead they write taglines like this like Kristoff wager Dave 25 years ago the researchers would learn how the brain achieves Consciousness by now um but the quest continues but this is actually not what the BET was about but that was about the neural correlates not the causal mechanisms not the generative production of Consciousness and remember not the same thing and there's always a relevant XKCD and so I direct you to this one here um but it gets worse so here's what science science had to say um Advocates of the losing idea are not conceding yet so again neither idea lost remember or rather both of them lost neither one of them won nobody lost nobody won um and so why would science say that one of them lost or is even losing it seems a little over the top and so it kind of makes you wonder are they like trying to drive clicks or what what is the point here of sensationalizing quite to this degree it seems like a bad idea it seems like oversimplification that actually changes the meaning of of what was presented so here's another example this is from The Economist um this article largely reported on a different event that was actually part of the assc like primary meeting but it still capitalized on the the buzz that this public event this evening event created um and it opens with this really kind of terrible Twist of the actual experimental design so the economist article says top of the bill was the announcement of the result of a so-called adversarial collaboration between proponents of two hypotheses about the nature of Consciousness again not what the BET was about it wasn't about the nature of Consciousness but anyway um this involved running a series of experiments begun in 2020 to determine which if either of them is correct again that's not what the collaboration was designed to do right in science you can never prove anything you can never show you're correct you can only show that you are not wrong or that you are wrong that's it that's what you get in science okay so moreover all of these articles the three that I've shown you and a number of others um they also contained essentially no discussion of whether these two theories Global neuronal workspace Theory and integrated information Theory one of which was losing apparently um whether they were the only ones worth considering so the Articles didn't really discuss anything else any other possible theories in fact the the Articles called these two theories leading by the media but the truth is that they're leading only in number of Publications not an amount of scientific evidence supporting them they just have a lot of Publications out there there's no discussion in these articles Beyond a trivial passing mention that there are any other theories out there at all and especially that they are not whether they

are viable or not whether they're empirically supported so here's some this isn't all of the empirically supported theories of Consciousness out there there's a really nice article here by Anil and Tim uh Anil Seth and Tim Bain that came out last year that discusses a bunch of these and there's a really nice article here um by this group also saying they actually looked at the empirical support for different theories of Consciousness how many papers are out there that actually challenge versus support each Theory and so this landscape is the one that should be the one to evaluate all of the results that were presented at this event in it means that even if one of the two theories that this one project was looking at if one of them didn't make the cut that doesn't mean the other one wins and is right it just means that one of them may be lost and it didn't even lose remember in fact so in addition to these really nice papers uh there's also a whole series of these adversarial collaborations from the same funding agency as the one that was shown in his public event that are designed to actually pit these theories against each other in a pairwise fashion in a like Thunderdome style for All You Mad Max fans out there I love old school sci-fi and the idea here is that two theories enter and one Theory leaves one Theory survives but it doesn't win because there's another round coming and so all of these nuances they were just unfortunately missing from all the news coverage and I understand that news likes to get people to read it and it likes to be Sensational but the missing aspects of these new the missing Nuance really changes the meaning of what was presented and that's actually a really big problem here uh and why is this such a problem you might ask I'm getting myself all worked out here right um so the answer to why I'm getting all worked up really comes back to what I presented in the beginning which is that we Consciousness scientists have this uphill battle we constantly have to fight to legitimize our research program and importantly this isn't just in the eyes of the public so it's not just in the eyes of the people reading these articles who are not part of this scientific sphere it's also in the eyes of the scientific Community more generally of our funding agencies and maybe most importantly our close peers within adjacent Fields so imagine for a second that you are on a hiring committee at a university and you read a candidate's description of their research program and it is about okay they have pages and pages and several Publications and they maybe have a grant that's about Consciousness science but you as the member of the hiring Committee just read this this New York Times article about how plants probably have consciousness and so maybe your eyes are now rolling out of your head because you've been primed to think this isn't real this is Sensational this isn't something that can be studied scientifically you might as well be talking about doing research on telepathy is this really a colleague that you want so here's another example what if you're

on a grant Review Committee or you're reviewing a paper and you just read these articles in science and in nature that heavily imply that there are only two real theories of Consciousness that have any empirical support but the paper or grant that you're reviewing is not targeting either one of these theories so now maybe you see the problem here these journalism articles they're they're not just infuriating they're really actively damaging the legitimacy of our field they're cutting us off at the knees in the eyes of our peers in terms of the financial support we can compete for from granting agencies in terms of where we can publish our science and by the way that science is state of the art and cutting edge we have a responsibility though as people who are doing the state-of-the-art Consciousness science to try to work with the media so that the reporting can be more honest so you know we can continue working there as well this is a two-way street in some ways but we also hope that folks who are listening right now will recognize this pattern the next time you read an article about Consciousness science then maybe those sensationalized claims are not exactly a reflection of the state of Art and in the field because here's the truth this is real Consciousness science modern Consciousness science it's a collection of basically every single state of the art discipline and method of scientific inquiry that you can think of so my version of Consciousness science that I do in my lab is a conglomeration of cognitive sciences and computational Neuroscience and a little bit of computer science artificial intelligence machine learning thrown in we also have heavy influences from philosophy and psychology and engineering and a little bit of meta science and we have neuroimaging we have statistics and this is really a representation of Consciousness science not just in my research group but the conscious of science being done worldwide by thousands of people conscious that science is right at the center right at the heart the very bleeding edge of our Collective scientific understanding um so remember when I put this slide up before and I said that the study of this question straddles the border between Natural Sciences and philosophy in a way that a lot of other domains maybe don't need to do and as a result it captures our imagination it gets out there it gets into your brain and your mind and sticks there in a way that some other fields might not but maybe you can see through what I've talked about today that the challenge is uh with communicating the Crux of this question and in doing the science itself that actually makes the people who are doing this pretty brave especially the younger trainees when you think about it they're fighting this real uphill battle so personally me I'm relatively safe I have job security if people don't take me seriously when I say I'm a conscious scientist that's their loss that's fine you know but the younger people the trainees in the field those who are trying to get jobs those who are trying to

to push the envelope of what we know about our brains and our minds and our awareness of ourselves in in this world they've got a lot of misperceptions to overcome uh so if you are excited and you want to know more about the modern exciting field of Consciousness science I suggest that you head to the assc website you can have a look at the whole Science Program they've actually posted videos of the Keynotes from this year so that's pretty cool you can go and see some of those um they can you can also check out the plan for the future um and of course if I've inspired you even a little bit to find out about Consciousness science yourself despite my kind of Grant here um I say welcome please come join us there's a lot of people doing some really cool stuff really amazing mind-blowing stuff working to figure this out so please come join us absolutely the more the merrier uh and so I think I will stop there and just say if you've enjoyed uh this talk or or anything that I said resonated with you I've got a real short little piece here it's about 800 words so not a long homework assignment you can go and check that out um so I think I'll stop and pass the mic back to Daniel and also welcome Nora Bradford who is a doctoral student here at UC Irvine who works in Consciousness science but who also is an accomplished science journalist and he's written articles in Scientific American Science News discover and many more so thanks very much awesome wow cool Dora baby to you for our first reflection about what what you see there or how you came to be here yeah sure um yeah so I was at assc and went to the event um that was associated with assc but not run by assc just to be clear um and it was really exciting to see Continental science kind of being celebrated in the way that it was I had some issues with the event but overall like I was very excited that it was getting this like public attention right um and then afterwards I was again like excited to see that people were writing about my field because that doesn't happen I'm like I could do um like immediately kind of disappointed by not wrong um yeah and I guess I started to think about kind of why conscious science is maybe a harder journalist to cover and I think there is just probably a lot more controversy and a lot more room for mistakes to be made um then perhaps in fields where the research is more clear-cut or more easily interpretable um so yeah I think there's a lot as journalists and also as researchers from this yeah I'll um can I can I just really probably follow up and say I I think that Nora you're spot on by saying that uh it's or by presenting the idea that it's it's the interpretation that's part of the problem here it's not just that people are maybe getting facts wrong when they're writing about this field it's that the interpretations are really nuanced they're really difficult to get right and it's really easy for an oversimplification to end up fundamentally changing the meaning in this space in a way that maybe doesn't occur quite as often in some other fields yeah that that's

really interesting to kind of disentangle the features that that make this a unique although also exemplary case of communicating challenging ideas I mean the first that presents itself to me is just how much is on the table I mean whose pain matters whose experiences or what experiences even matter what's the difference between a who and a what this is the real question if something doesn't have an experience then there's no feeling but if something does have an experience then it's it's a feeling and it's a morally um culpable or um something that we're responsible for treating in an ethical way so it's like whether or not it's always brought up in those articles the stakes are so high whereas whether a given molecular compound does this molecular interaction is closer to the objective whereas this is the ultimate subjective questions seemingly sure yeah I I agree with you definitely that the ethical Stakes the the moral implications of this field are are just staggering and certainly there it's not unique in that that way that the ethical and moral implications of sociological research or climate science or you know of many other domains also have like these staggering implications um I think one of the the biggest challenges that we have is not just in the interpretation of objective data but exactly as you've just said this is the ultimate subjectivity that the thing that we are studying by its definition is not objectively observable and that is absolutely something that is uh at least you can design measurement Tools in a way that have different philosophical foundations in basically every other field but by its very Nature by the definition of subjective experience we cannot design a measurement tool that actually says what is it like to be you that's kind of the whole problem but your your uh your points also about the ethical implications in medicine the difference between a who and a what um I think that that's also really important to bring it outside here we are trying to to you know I presented this as like we want to understand Consciousness and I never really defined consciousness he just said like here's Consciousness and like here's an umbrella term and you you take that kind of how you want um but it's true that Consciousness science spans all the way from you know wakefulness versus coma uh or locked in syndrome or traumatic brain injury um anesthesia all the way through to uh is a baby conscious is a fetus conscious is an AI conscious is a cockroach all right or your dog uh or you know 200 years ago people might argue that children were less conscious than adults and so it was okay to hit your children because they wouldn't feel it as much or something and so there's there's a lot of sociological and ethical and moral implications in everything that we're doing which is why one of the reasons I've chosen this field too is that it really scratches a lot of intellectual itch but it also feels really important especially right now with the rise of machine intelligence and with uh ongoing debates about

animal uh or fetal Consciousness that really bear extreme impact on legislation on you know moral decisions that we make as a people so yeah it's it's a big thing that we're doing I'm a little biased but it is a big thing and I would add that even if researchers are sometimes good at explaining what they mean by Consciousness in their research like two journalists say the journalists are not always necessarily great at communicating that to the public for their writing um and I get why that would be hard it's kind of it can be fun it can be like awkward and unappealing to explain but that is crucial because people readers can take it however they want based on how they're interpreting the word Consciousness um so I think that's super important to address if you're writing about consciousness yeah it's really a field that's kind of a leading indicator because especially if approached in a scientific way in our current setting uh a scientific understanding of X is what shapes evidence-driven policy about X so I guess this leads to what what is or or would or could be a science-driven approach like how do we know it when we see it when someone is approaching this question from a scientific perspective what would each of you say to well I'm going to give kind of the stock answer which is that you want to have objective measures that are valid and reliable um but the the big one is falsifiability and that's you know any student of science going back to grade school is going to learn a little bit about that which is that if you've got a theory or you've got a hypothesis uh about how let's say the brain uh generates Consciousness or even going back to just correlates just like which bits of the brain are correlated in their activity with whether you're more conscious or less conscious so let's keep it really like very basic um you need to have it be falsifiable you need to have say it's this area of the brain that will correlate and not that area of the brain or that uh and the recognition that an absence of evidence is not evidence of absence is also really crucial to this idea which is that just because I failed to measure a correspondence between the activity and the bit of the brain that I hypothesized and you know some measure of awareness that I get from self-report doesn't mean that that area of the brain isn't doing the work it just means that our machines are terrible or that our measure sucks um and and honestly despite the state-of-the-art measurements that we do have we have you know millimeter level Precision in humans we have cellular level precision and non-human animals we can record a lot of stuff and we have a lot of really fancy machine learning algorithms to help us extract what that stuff means to go from super high dimensional noisy data down to something that we can maybe make the argument is interpretable uh inter interpretable yeah um but just because you didn't find what you were looking for doesn't mean that it's not there it just means that you were maybe looking in the wrong place or that you were looking using

the wrong tool uh and so these these kind of go hand in hand because on the one hand you want your hypothesis to be falsifiable so you aren't allowed to just say oh well I just wasn't looking for it the right way so I'm just gonna you know revise how I'm looking and try again and therefore I'm still right you need to design your experiments and your theories such that if you find a particular result a positive result that it actually disproves your theory as opposed to just failing to provide support for your theory um so I think that those those those and you know other kind of standard criteria that you might find in any stem classroom that demonstrate the scientific method uh those are probably the the definitions that I would fall back on and I think that Consciousness and science broadly speaking the stuff that is presented in assc it does all of that um it absolutely does it's the same caliber of computational Neuroscience and neuroimaging and psychological psychophysics measurement and logical philosophy approaches that you would find at any other Flagship conference in the in the world yeah and I would say if you're reading um an article rather than like a journal paper like if you're reading something in the in the Press um the way to know if it's trustworthy or not or one way is if it's making Sensational claims like claiming to have you know found the root of Consciousness or some really large claim it's probably not true like I think we would that wouldn't happen in one study I'm assuming um so just read with a skeptical eye and um if you feel comfortable try to read the primary Literature Like get inspired by the the Press but if you feel like there's something off just go to the primary literature and see what's up and and see if you can interpret for yourself it's understandable if you can but yeah that's what I would say that's a good rule of thumb Nora that like one study is never going to be the answer yeah not just ours but it's it's easy to get swept away in The Sensational claims and to be like ooh they found like the neural cause of Consciousness like probably not yeah yeah and I would say like pre-registration also I mean Megan I don't know like what your rule of thumb is for this but those I mean if people pre-register there um you know plan for the study and how they're going to analyze it ahead of time that is usually a really good sign I mean basically what you're just saying but just like stamp of we promise we did this ahead of time kind of thing um yeah avoids parking lotsizing after the fact right yeah yeah it was kind of cool to see Meta and also maybe kind of a bibliographic analyzes looking at different Trends in literature but then also going beyond the peer-reviewed literature to looking at the real trends that that influence people who are outside of the profession and it's like um it's kind of I don't know if Canary in the coal mine is the right um birds in the right Niche here but it's like um proposing unobservables is commonplace in research like latent cause analysis and and any kind of

structural equation modeling and here is the um the tour de force of unobservables maybe that which is defined as unobservables and so it's actually like a space to to bring up all these pieces of epistemic hygiene about from pre-registration to the PACE and the tone of the scientific community and questions of like multi-disciplinarity in research it's it's kind of um it's a fun and then there's a heart at the intersection of the Venn diagram where it kind of all comes together it's like well this synthesis cannot happen without all these pieces if we're not having philosophers in conversation with the statisticians then this synthesis can't happen now maybe the synthesis that you care about doesn't need that but we care about this and this synthesis requires it so we need to develop the approaches and the communication around that and um so it's uh really a tip of the spear effort even apart from the consequences of the research what a heartwarming sentiment thank you for sharing that I I think I love the phrase epistemic hygiene because that is so crucial I think to every field but it's like particularly aggressively exposed in this field which is exactly why I think that my science has benefited so powerfully from close collaboration with philosophers and with statisticians I mean this is kind of the space that I I live in here in the cognitive Sciences department is where we're halfway between Ai and philosophy and Psychology and Neuroscience and engineering and it's like it it's kind of like where all these things come together but then this meta science thing I'm happy that you picked up on that um because I think that it's also really the the bibliometric analyzes are really critical and we're at this crazy spot and have been for a long time now but I think it's really really coming to a head where the vast majority of our scientific knowledge is not accessible to anyone researchers simply because there's so much of it right you can't read all of the literature you can't even read all the literature that's in your tiny little corner and Consciousness science like many other fields in this way actually is one of those that brings together a lot of different fields so it makes it even harder for you to keep up with the things that are going to be really relevant I mean you have to keep up with The Cutting Edge tools in neuroimaging data analysis plus the epistemic hygiene that the philosophers are going to bring plus like Cutting Edge behavioral paradigms and what they mean and like you know naturalistic data uh collection settings so that you're not just saying oh I've discovered something fundamental about Consciousness by showing you Gabor patches all day and I say this as someone who shows people Gabor patches all day but it's it's really like it's a real challenge to bring together all these things and maybe you're right maybe that Consciousness science is I don't know about the canary in the cold mine so to speak but maybe you're right it's one of the places where you can really see all these cogs working

together to come at one major question or maybe you don't quite see the need for that hygiene and some other fields yet it still is there like latent cause analysis is not just something that we do in Consciousness science or Neuroscience we do it in economics we do it in sociology we do it like everywhere and so having a real handle on how to design stuff that actually asks the questions that you're interested in is something that I find that I have to continuously work hard to keep in my sights um and and I'm lucky that we've got a pile of philosophers that we get to hang out with who blow into your talk and say everything that you just said is wrong and here's why here's this logical hole in what you've done and they're right and it's great because it makes the science better and you don't typically get philosophers you know blowing into a talk in a lot of other fields it doesn't happen and um and I think it should we all have that inroad to our experience so it is something that integrates across disciplines kind of even casts the disciplinary organizational Paradigm into um at least critical light and and then um goes even outside of Academia because these are these are what people search engine for and uh language model discuss and I think it is one of these um it's like an itch that some can't scratch with this inscrutable and ineffable nature that that does Verge into the mystical and then like so what do we do with the huge mystery and and so that that makes it a um a a training and exhibition fields um again that's not just important but like is kind of the the the archetype of well how do you have a multi-decade conversation with potentially non-compatible theorizing whereas in many of the physical sciences including physical reductionist perspectives on on neuroscience threads open and close a little bit quicker a little bit easier tend to not um delve into ancient history tend to not need to invoke World Knowledge Traditions so that makes it fun it makes it fun and useful and if something's fun and useful what what secret third thing are we looking for exactly sounds like uh if you weren't already a Consciousness scientist maybe we've converted you um as as only a bug doctor I did did uh one one work on the uh ant colony and definitions of Consciousness and we didn't come down on the issue of whether like the nest mates are conscious under what definition or the colony um but more pointed out well if you define it according to some neuro mammalian neuroanatomy then you kind of already know what the answer is going to be it's going to be no and if you define it according to a specific um primate censored symbolic communication approach again you know what that's going to be it's no if you define it based upon some generic informational property then you already know what it's going to be it's going to be on that continuum and so we kind of tried to flip the approach by instead of asking like in which teacher would help us differentiate whether the ant colony is conscious or not we just said let's just look at what the features that

people are using and then ask whether colony is but but we don't have we don't have the um one meter metal bar for Consciousness so again it's like well let's say 7 out of 10 that we observed we don't know how many other unproposed theories there are so 7 out of 10 could be total um uh sampling bias um even if it was 10 or 10 then what would that type of consensus really mean and um that's where our road took us and I I saw that conclusion as like a chance to broaden the discussion but discussions are not only about broadening because there's real decisions that are being made so it can't only be yes and and I think that when you connect it back to falsification and all of these other properties that's where we get the the winnowing but it's those two complementary conversations like the broader um inclusion of perspectives that expands the space that we can comprehend and then the diligence that that also shrinks it back down and somehow will zigzag to better and better understandings your uh your point about we don't have a yardstick we don't have a metal rod right that says you know how conscious I mean like according to to some folks we we have that metal rod and we've surpassed that with for example large language models uh they're uh and we're not just talking you know particular Google Engineers who decide to get really excited about how their pad language model gets is conscious we're talking any any kind of reasonable interpretation of the old school Turing test would be passed by existing language models right now in fact arguably it might have been passed long ago with Eliza uh in like the 70s um and and if it wasn't passed with Eliza then certainly it has been passed now but I think that there is a growing understanding that information processing capacity and intelligence are not the same thing as internal phenomenal experience and that the tests that we have to the extent that we have any tests at all are willfully inadequate and as you said are locked to particular implementations so you can't apply the mammalian test for Consciousness like oh I'm looking for this particular neural signature which tells me whether the monkey is anesthetized or not I can't apply that to ants and expect it to work it's not going to tell me anything useful there can't necessarily apply it to octopuses either for the record so we're not even going insects we're going other arguably incredibly intelligent beings that can open jars and be sneaky and like do all sorts of cool stuff and uh and that have complex social interactions and are basically aliens this is my my you know professional scientific opinion here what they basically are right like they're so fundamentally physically different from us and so any consciousness of meter that we develop for us or that applies in monkeys or even in dogs or cats is not really going to apply there um and so differentiating tests for intelligence from tests from like inner experience it's a really difficult problem and then there's the third problem which is a test for whether we attribute

Consciousness to an agent which if you saw a movie X Machina is the Garland um so that's like you know do never mind whether it actually has Consciousness do I attribute Consciousness to it because that is going to change how we interact with it and the rights that we ascribe to it and so on and so I think we're well past the point at which we need to get a better handle on this some of the stuff that I'm working on right now is moving in this direction not just me and a huge group of lots of other people um and I think Joshua bengio gave testimony to Congress like today uh about the need for uh increased regulatory oversight of uh who has access to so-called artificial intelligence models so that we can better understand how their implementation and their interaction with humans might exacerbate biases or cause all sorts of other problems um so you can actually go and read his his testimony that he gave I don't remember whether the link is you can find it um but he had a summary and then he links to the testimony and I think that having folks like yahshua and other Giants in the field speaking out about the need for actually tackling this and not this isn't just a philosopher's playground to talk about Consciousness this is this is as you said it's a testing bed for where the rest of science is going and it's also like very topical right now so what is that healthy conversation looked like in the case of the synthetic intelligences and language models and all of this what what does that continue to look like between the actual measures and proxies that we use in research and uh I mean what memes or themes or words that I guess by definition aren't popular today like big data wasn't Popular until it was so what some something like what has to come on the scene and come into the lab do you think to support the kind of um like work and knowledge that that either of you would want to see thank you this isn't an answer to your question but it just reminded me of one thing that we need less of is people writing about these interactions with AI that are like my AI told me that it wanted to get married or like my AI whatever like that's not it's not useful it's not news stop reporting about it that's all I have to say okay go ahead no I I wanted you to take this too because I think that the question is about Communication in a way that is you're you're much more embedded in the science communication space than I am and so for I think one possible way to go about this that maybe you can speak to would be what kinds of you know you're Daniel you're talking about what do we need to bring into the lab and maybe communication tools is one of those or ways of communicating this science ways of of end to different audiences um and so maybe Nora you've got some ideas even if you've done so much in this space I don't know yeah I mean I guess I what we've been talking about with people who study human consciousness people who study Consciousness and other entities as well like just

being clear about what you mean by Consciousness like do you mean a subjective experience do you mean um you know basically is a light on or off like uh do you mean awake versus asleep just being really clear about what level you're talking at I think is really helpful again when you're just like presenting your research whether it's on Twitter or you're talking to a journalist um really being clear-cut about that so that people can't make kind of extrapolations from your work that your work doesn't cover I don't know what else I mean yeah I guess just being willing to also talk about your work I feel like a lot of data science feels pretty inaccessible to the general public um so finding ways to communicate what you do and how you're conducting your work is helpful being open about it yeah there's a lot of jargon in our field too in any field there's jargon um but the the oversimplification this is going back to what we talked about at the beginning the oversimplification of jargoning terms in our field seems particularly susceptible to creating a situation where the the facts haven't changed but the interpretation has uh and so I think it's almost like sensitivity training for anybody who's in this space like just kind of a you know a little a tutorial that you can do that's like a two hour you know click-through thing it's like here are all the ways in which communicating this type of science due to its nature is more challenging than you might anticipate and so my my own involvement in um in a program run by the Canadian Institute for advanced research has given me a little bit of training and speaking with the media and I would never never say that I am an expert in this at all uh I will only say that uh it was eye-opening and it was very useful and the clarity of language that you are used to using the the Precision of language that you're used to using in a close scientific sphere versus when you're speaking to a more General audience and when you're speaking to a journalist or with several journalists who are then going to be writing for a more General audience it's a it's a different mindset it's a different way of speaking and even just being aware of that I think is really really useful it's not that one is better than the other it's just that they are very different and that when we are training to be scientists when we're doing a PhD in a field related to this speaking to the general public is not a skill that gets pressed on it's not a skill that gets it's part of like your professional development training maybe you do like one practice talk somewhere but it's not typically part of a general training protocol and So speaking to non-specialists speaking to the media is speaking to high school audiences speaking to your mom at the dinner table like learning how to do this effectively can actually be really valuable and really rewarding and really powerful and again I would never say that I am an expert at this at this point I just like to think that I'm a little better at it than I otherwise would be had I not had it drilled into me how careful I need to

be I would also say um don't be scared to like email journalists if you see that something's inaccurate or you know speak up like Megan has done um about the kind of how things are portrayed inaccurately because yeah as a journalist like no one's ever done that to me but it would if something wasn't accurate I would want to know and I would want to fix it like ASAP so don't feel bad about emailing journalists they want their pieces to be accurate as well that's what I would add that can add to the conversation yeah it does make me feel a little bad about like you know calling out the pieces that I did in in the presentation in the the piece of writing that I did too but at the same time like they put this in writing and this is wrong and and it's I I don't like making people upset and I don't like making people angry it's out there in the world and I saw a million eyeballs and that's a problem and so we're hoping your viewership is about a million right Daniel so give or take and so no Brilliance for everybody so this will cover everybody who read those those articles right it's it's interesting like the field is about awareness of other Minds our own mind but also awareness of other minds and in theory but then in practice awareness of other Minds is communication because if we say something that it's like well I wouldn't be annoyed by that or I wouldn't be angry with that but it's like but you missed the point of communication which it's not about what you would think about it it's about because the recipient would be so it's kind of like the theory and the practice of of um situational awareness about different Minds all of which are different than our own some are potentially radically different than our own others might be more similar um so it's there's really um maybe even a quintessential relationship between Consciousness studies and communication studies which provides like a really unique two-way street that that um the world is swerving into that to Lane Highway with these questions that are grabbing people's salience what what are your sort of um like closing thoughts or directions or what are some exciting areas that that you'd like to see unfolding in the coming months and years well I'll go first yeah first of all I I think I communicating conscious to science being about knowing about other Minds I mean it's like effective communication is theory of mind and theory of mine may have a lot to do with modeling other minds and modeling your own mind and yes absolutely it's all it's all wrapped up I think for me um getting this right going forward or continuing to to work towards getting it right is a priority for me uh and one of the the pieces scientifically that I'm working on that I hope to then communicate effectively when we get it right or several of the pieces is I have a couple um these really wonderful large collaborative projects where we're working on everything from figuring out how you would actually measure of Consciousness effectively or what it might even take to measure Consciousness

effectively in other entities that are not necessarily like awake behaving and healthy adult human beings uh all the way to Let's actually design some experiments to test several different theories of Consciousness let's design a new uh some of what Nora's working on actually is designing new measurement approaches to say well we need to actually like have a quantitative handle on the things that we're talking about so this becomes objective it becomes falsifiable and becomes like we can do model arbitration so all these things I'm really excited about but overarching all of those is this idea of going it's all connected right you've talked about how Consciousness science might be this this Leading Edge or wave front of how uh interdisciplinary science and meta science needs to be done going forward as uh as we become even more Theory driven in some of our uh scientific lines of inquiry than we have been in the past and more interested in latent variables and causal Discovery and so on and so forth and as the science teams get bigger and more complicated and the jobs within those science teams become more specialized and we have more and more need for interdisciplinary collaboration we're gonna have to change How We Do team science and how we do credit assignment within team science in a way that I think might end up being kind of incompatible with the nice siled cute departmental structure that we have in our universities uh and so a focus on not just tolerating but celebrating interdisciplinarity celebrating team science doing credit assignment correctly and you know saying first author isn't the only thing that matters that like it really matters if you wrote this gorgeous software package that facilitated a whole bunch of stuff like that's also a meaningful and Powerful contribution even if you didn't get first author on that paper you still deserve a lot of credit so I think that there's a lot of shaking up that we need to do about how we do science going forward and I really like the idea of using Consciousness science as a case study I think that's beautiful and thank you very much for making that lovely connection because it really locks together a lot of the things that I've been thinking about recently as well so there's my my parting it's a yeah I was also going to bring up the these huge projects that you're a part of and that other people are part of it's really exciting to see so many people working together in these huge teams I feel like it just makes science so much more efficient and more careful so that's very cool another thing I'm hoping is that one the journalists don't get scared of covering Consciousness science from this I think like we'll learn from our mistakes and we'll move forward I I don't think we'll get scared but I hope that there's more coverage of this field um and I also hope that I mean something that I've noticed about science journalism is they really like to just have clear-cut findings because that's the simplest thing to communicate to people but I think we can as a society like learn to

be comfortable with the ambiguity a little bit more um especially when it comes to conscious science like we might have found something but we're not sure and that's okay and that's science like it's okay if your headline isn't Snappy and clear it's important for it to be clear but snappy and I guess have one takeaway point it's okay if it doesn't have a takeaway point um yeah I think that's what I'm excited New Era of science journalism well it sounds like bigger teams bigger Community more Theory more data more philosophy more ambiguity it'll be a great time more philosophy more ambiguity love it yes thank you Megan and Nora for joining always um welcome back so till next time yeah thank you so much this has been really a pleasure thanks for having us thank you bye [Music]